Paritosh Singh

+91 8755814015 | paritosh0180@gmail.com | Bengaluru, India
linkedin.com/in/paritosh18 | github.com/Paritoshcodes | paritoshsingh.me

Education

SRM Institute of Science and Technology, Chennai Jun 2023 to May 2027
B.Tech, Computer Science and Engineering (Data Science) *CGPA 8.78/10; Class XII 76.6%; Class X 84.4%*

Experience

Software Development Engineer in Test Intern Jun 2026 to Present
HealthEdge *Bengaluru, India*

- Roughly doubled the number of failing tests the team triages per day by building a Python tool that parses failed Katalon runs, sorts them into 15 failure signatures across 7 categories and proposes a fix for each.
- Kept every proposed fix traceable to existing code by adding retrieval (RAG) over the repository's locators, custom keywords and past failure logs, then rerunning each fix headless so regressions surface before code review.
- Blocked hallucinated test code at the source by building a TypeScript MCP server that converts Zephyr test cases into Katalon tests and rejects any method, locator or value not already present in the repository.
- Merged 150+ automated UI regression scripts in Groovy for a healthcare payer web application, each through code review in two week Agile sprints with developers, QA and the test lead.

Projects

mini-redis | *C++, POSIX sockets, epoll, Linux* github.com/Paritoshcodes/mini-redis

- Sustained 150K+ SET and GET requests per second across 50 concurrent clients on a single core by building a single threaded, non blocking epoll event loop over raw POSIX sockets.
- Scaled to 2.7M+ GET and 2.4M+ SET requests per second at 128 deep pipelining by writing a RESP parser that buffers partial frames, so batched commands survive split TCP reads.
- Shipped 18 Redis compatible commands, including SET with EX/PX, TTL, INCR and pub/sub (1,000 of 1,000 messages delivered under test), in a 594 line C++ server driven by stock redis-cli and redis-benchmark.

Revenant | *Python, FastAPI, TypeScript, PostgreSQL, Playwright* github.com/Paritoshcodes/revenant

- Measured a payment recovery agent's causal lift, not gross recovery, with a stratified 50/50 randomised holdout and bootstrap intervals, reaching 93.8% coverage of a known true lift over 2,000 simulated runs.
- Found and fixed a bootstrap interval that was 38% too wide by building a calibration harness that reruns the estimator against a world whose true lift is frozen in code before any experiment runs.
- Closed a prompt injection path found in adversarial testing by restricting the LLM to returning a broad failure class, which trusted code maps onto a fixed grid of retry actions.

Agent Replay | *Python, AWS Bedrock, Strands Agents, Lambda, DynamoDB* github.com/Paritoshcodes/agent-replay

- Built at the WeMakeDevs x AWS hackathon: localised agent regressions to the exact failing step by deriving a YAML contract of required tool calls and their order from 5 recorded runs, judged by set logic, not an LLM.
- Gated every pull request on that contract through a GitHub Actions workflow that assumes an AWS role via OIDC, keeping zero long lived AWS keys in the repository or its secrets.

Vigil | *Python, FastAPI, PostgreSQL, XGBoost, LightGBM, SHAP, React, TypeScript* github.com/Paritoshcodes/Vigil

- Forecast seven natural hazard types across 25 Indian cities at ROC AUC 0.74 to 0.88 (heatwave 0.876, cyclone 0.852) by training XGBoost and LightGBM ensembles with SHAP explanations behind a FastAPI and PostgreSQL backend.

Open Source Contributions

Martin | *Rust, MapLibre vector tile server, 3.9K GitHub stars* github.com/maplibre/martin/pull/2599

- Fixed a redirect that ignored `--route-prefix` for .pbf tile requests; merged in PR #2599 with a regression test and shipped in release v1.10.1.

render-engine-cli | *Python, static site generator CLI* github.com/render-engine/render-engine-cli/pull/40

- Fixed broken sitemap URLs when serving sites locally; merged in PR #40 after two rounds of maintainer code review.

Technical Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, Groovy
Generative AI: RAG, LLM agents, multi agent workflows, MCP servers, prompt engineering, AWS Bedrock, Strands Agents
Machine Learning: scikit-learn, XGBoost, LightGBM, SHAP, pandas, NumPy, NLP (TF IDF text classification)
Backend and Databases: REST APIs, Node.js, Express, FastAPI, SQLAlchemy, JWT, OAuth 2.0, PostgreSQL, MongoDB, Redis, Bull, Socket.IO
Testing and DevOps: Katalon Studio, Playwright, Selenium, JUnit, Jest, pytest, Postman, Git, GitHub Actions, Docker, Linux, AWS (Lambda, DynamoDB, S3)

Publications and Achievements

Publication: Vigil multi hazard prediction paper accepted in ISJEM
Certification: SAP Certified Back End Developer, ABAP Cloud 2026
Certification: NPTEL Introduction to Machine Learning, IIT Madras, Elite grade 2024
Achievement: 2nd place, Capture the Flag, SRMIST 2024
Leadership: Associate Corporate Lead, SRMKZILLA (SRMIST's Mozilla club), ran sponsorships and events 2024 to 2025